

Deepak Datta Anguru

Ph.D. Research Scholar

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About

I am a Graduate Research Scholar Working in the Analysis, Modelling, & Design of High-Performance Power Electronics

Education

- 1.2025-current **Direct Ph.D. Power Electronics and Motor Drives at IIT Madras, IN**
 - **Research Area:** Modular Charging Infrastructure for High-Power EV Fast Charging. Supervised by Prof. Deepak Ronanki.
 - M.S(R) + Ph.D. (CGPA: 9.02)
- 1.2021-4.2024 **B.Tech. Electrical and Electronics Engineering at JNTU Kakinada, IN**
 - **Thesis:** Reconfigurable Wide Voltage Range Battery Charger for Electric Transportation Applications. (Major in EEE, Minor in CSE)
 - Graduated: First Class with Distinction. (CGPA: 8.93)

Appointments

- 1.2025-current **Research Scholar, IIT Madras, IN.**
 - Working with Prof. Deepak Ronanki at the IITM-PEEM Lab on Modular Charging Infrastructure for High-Power EV Fast Charging.
- 6.2024-12.2024 **Project Associate, IIT Madras, IN.**
 - Working with Prof. Deepak Ronanki at the IITM-PEEM Lab on Design, Modelling, and Control of Segmented Switched Reluctance Motor (SSRM) for Two-Wheeler EVs.
- 2.2024-5.2024 **Research Intern, IIT Madras, IN.**
 - Working with Prof. Deepak Ronanki at the IITM-PEEM Lab on Design, Modelling, and Control of Reconfigurable Power Converter Topology for Wide-Voltage Range Applications.

Publications

Patents

- [P1] D. Ronanki and **D. D. Anguru**, "A Method and Control Framework for MMC Submodule Open Circuit Fault Diagnosis with Submodule Redundancy," IPO Appl. No. 202541066960, Jan 16, 2026. [Filed]

IEEE Peer-Reviewed Journal Articles

- [J1] **D. D. Anguru** and D. Ronanki, "Pantograph Charging for Electric Vehicles: A Comprehensive Review: Enabling efficient and high-power fast charging for buses, trucks, and heavy-duty electric vehicles," *IEEE Electrification Magazine*, vol. 14, no. 1, pp. 66-78, March 2026. [↗]

IEEE Peer-Reviewed Conference Proceedings

- [C1] **D. D. Anguru**, B. N. S. S. Vempali, D. Ronanki, and S. Srinivas, "Submodule Redundancy Based Fault-Tolerant Control for Modular Multilevel Converters: A Comprehensive Review," in *Proc. IEEE 6th International Conference on Sustainable Energy and Future Electric Transportation (SEFET)*, Nagpur, India, 2026.
- [C2] **D. D. Anguru** and D. Ronanki, "A New Multilevel and Multiport Topology for Electric Vehicle Fast Charging Applications," in *Proc. IEEE 6th International Conference on Sustainable Energy and Future Electric Transportation (SEFET)*, Nagpur, India, 2026.
- [C3] **D. D. Anguru**, J. R. Rahul, and C. Vadrevu, "Reconfigurable Wide Voltage Range Battery Charger for Electric Transportation Applications," in *Proc. IEEE International Conference on Power Electronics, Drives and Energy Systems (PEDES)*, Mangalore, India, 2024. [↗]

Achievements

2026	Recipient of the IEEE IAS Student Travel Award for the IEEE SeFeT 2026.
2024	One of the Best Presented Abstracts at ISRO National Space Science Symposium .
2023	1 out of 80 selected for ISRO SSA-STM Workshop conducted at ISRO HQ, Blr.
2022	Project titled "Unique Health Card" (UHC) selected to final rounds of Smart India Hackathon and YUKTI-National Innovation Repository conducted by the GOI.

Professional Activities

2026-current	Reviewer for <i>IEEE Transactions on Transportation Electrification</i> .
2024	Reviewer for <i>Int. Conf. on Power Electronics, Drives, and Energy Systems</i> .
2025	Reviewer for <i>IEEE Energy Conversion Congress & Expo, Asia</i> .

Technical Societies and Committees

2024-current	IEEE, Graduate Student Member	2021-2024	IEI, Student Member
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Skills

Programming	C, Python, LaTeX, MATLAB
Software	Altium PCB, Ansys Maxwell, Simulink, Altair PSIM, LTSpice, PLECS
Tools	Opal-RT HIL, dSpace, Microsoft Visio, Inkscape, Microsoft Office
Web Tech.	HTML, CSS, JavaScript

Languages

Telugu	Mother Tongue
English	Working Proficiency
Hindi	Conversant
German	Elementary Proficiency
Russian	Elementary Proficiency